

DHANVARDINI RAJENDRAN

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SUMMARY

AI Data Engineer with 3+ years building end to end systems on **Azure Databricks** and **Snowflake**, from **ETL/ELT** pipelines to **FastAPI** and **Django** services and **RAG** and **agentic AI** applications shipped to production.

WORK EXPERIENCE

Optum, UnitedHealth Group

Software Engineer (Data Engineering and Analytics Platform)

Feb 2023 to Aug 2024

- Delivered **10+ production ETL pipelines** on **Azure Databricks** processing **4B+ pharmacy claims records**, publishing Gold layer **Delta** tables consumed by 5+ downstream teams.
- Productionized a **pharmaceutical supply chain intelligence platform**, engineering 5+ production grade **PySpark** pipelines from a proof of concept.
- Built **FastAPI** and **Django** services and **React** interfaces exposing curated datasets and pipeline status, replacing manual pulls for teams supporting **5M+ health plan members**.
- Mentored 70+ new joiners** onto the platform and cut pipeline failure rate from **40% to 10%** through root cause analysis and multi layered **data quality** gates.

Associate Software Engineer (Data Engineering, Platform Reliability)

July 2022 to Feb 2023

- Enhanced a multi step execution framework in **Python** and **PySpark** with reconciliation, deduplication, and rerun from failed step recovery.
- Deployed data assets across Azure Databricks workspaces through **Unity Catalog** with fine grained access control and cross workspace lineage.
- Implemented **SQL** performance optimizations including predicate pushdown and selective aggregation on billion row **Delta Lake** datasets.

PROJECTS

PodcastIQ: Snowflake Native Agentic Search Platform

Mar 2026

- Architected a 4 layer **Snowflake** warehouse with **dbt Cloud** and **GitHub Actions CI/CD** powering an **agentic** semantic search agent with query routing, reaching 95.8% router accuracy at \$0.0012 per query.

IMDb Analytics: AWS Cloud Native Warehouse

Dec 2025

- Built an **AWS** cloud native ETL over **190M rows** from 7 datasets, landing S3 to **Glue** and modeling a **Redshift Serverless** star schema with 5 conformed dimensions, tuned with **DISTKEY** and **SORTKEY** and served through QuickSight.

DocuParse: SEC Document Intelligence Pipeline

Jul 2025

- Built a **DVC** orchestrated ML pipeline parsing SEC filings with **Detectron2** and **LayoutLMv3**, extracting **187 tables** and **2,954 XBRL** concepts at **99.56% accuracy**.

META TradePulse: Real Time Trading Signal Platform

Apr 2025

- Engineered a **Kafka** and **PySpark** streaming pipeline computing technical indicators, persisted to **Snowflake** through Streams & Tasks for sub minute refresh.

EDUCATION

Northeastern University, College of Engineering

Aug 2024 to May 2026

Master of Science in Information Systems (GPA: 3.7/4.0)

Courses: Generative AI with LLMs, Big Data Systems and Intelligent Analytics, Data Architecture for BI, Database Design

SASTRA University

Jun 2018 to Jun 2022

Bachelor of Technology in Computer Science

CERTIFICATIONS AND RECOGNITION

- AWS Certified Data Engineer** (Mar 2026), **AWS Certified Solutions Architect** (Jan 2026), **Microsoft Certified: Azure Data Fundamentals** (DP 900).
- Laurel and Scroll 100**: top 100 students across all Northeastern campuses for leadership and academic excellence.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, JavaScript, Java, FastAPI, Django, Node.js, Express, REST, Redis

Data Engineering & Streaming: PySpark, dbt, Azure Data Factory, Azure Databricks, Snowflake, Delta Lake, Kafka, Airflow, DVC, GitHub Actions CI/CD

AI: LLM and RAG pipelines, agentic workflows, Snowflake Cortex, LangChain, LlamaIndex, embeddings, Pinecone, Claude

Databases & Apps: SQL Server, PostgreSQL, MongoDB, Neo4j, React, Next.js, Streamlit, Plotly

Cloud & BI: AWS (S3, Glue, Redshift, QuickSight), GCP BigQuery, Power BI, Tableau